Coalition Theory

Master Equation Index

Textbook-Format Mathematical Review

This is a visual review copy of the current Master Equation Index ledger. Equations are typeset in standard mathematical notation for human review. No scientific reconciliation or equation redesign has been performed. Any rendering-only syntax repair required to overcome a spreadsheet/OOXML artifact is documented at the end.

EQ-001 — Transmissivity Framework Function

$$T = f(\rho_m, \rho_E, \sigma, \theta, \Phi, \mathbf{B}, \mathbf{E})$$

Operational context: Corridor definition evaluating localized field environments
Primary simulation reference: SIM-CORE

EQ-002 — Transmissivity Saturation Condition

$$\frac{\partial T}{\partial t} \to 0$$

Operational context: Indicates static boundary state stabilization
Primary simulation reference: SIM-CORE

EQ-003 — Transmissivity Spatial Gradient Breach Threshold

$$\nabla T > \Gamma_{textbreach}$$

Operational context: Defines boundary breakdown point where containment slips
Primary simulation reference: SIM-CORE

EQ-004 — Permeability-Transmissivity Scaling Constant Matrix

$$\alpha \cdot T \approx \kappa$$

Operational context: Couples medium conductivity directly to local transport profiles
Primary simulation reference: SIM-CORE

EQ-005 — Saturation Ratio Formulation

$$S = \frac{E_{meas}}{E_{ref}}$$

Operational context: Normalizes local measured energy field against reference vacuum states
Primary simulation reference: SIM-CORE

EQ-006 — Recovery Gate Threshold Condition

$$\frac{dS}{dt} > \Lambda_{rec}$$

Operational context: Dictates the timeline rate needed for system recovery to baseline
Primary simulation reference: SIM-CORE

EQ-011 — Integrated Corridor Path Resistance

$$R_\gamma = \int_{\mathcal{C}} w(s)\, \chi_{textcorridor}(s)\, ds$$

Operational context: Calculates net optical/field attenuation along a line element s
Primary simulation reference: SIM-PLASMA-01

EQ-012 — Net Boundary Poynting Flux Projection

$$\mathcal{P}_\gamma = \int (\mathbf{E} \times \mathbf{B}) \cdot d\ell$$

Operational context: Evaluates electromagnetic power flow intersecting containment edges
Primary simulation reference: SIM-PLASMA-01

EQ-013 — Local Coordinate Clock Synchronization Factor

$$\frac{dt_{obs}}{dt_{ref}} \approx 1$$

Operational context: Enforces non-dilated baseline clock frames in flat reference space
Primary simulation reference: SIM-PLASMA-01

EQ-014 — Spatial Metric Longitudinal Length Matching

$$L_{meas} \approx L_{ref}$$

Operational context: Ensures length consistency before severe localized metric squeezing
Primary simulation reference: SIM-PLASMA-01

EQ-015 — 6D Phase Multi-Vector Tracking Array

$$\mathbf{V} = [\Delta E, \Delta q, \Delta s, \Delta \mathbf{p}, \Delta B, \Delta L]$$

Operational context: Monitors instantaneous updates across primary thermodynamic variables
Primary simulation reference: SIM-PLASMA-02

EQ-016 — Environmental State Distribution Matrix

$$\mathcal{D}_i : P_i(textenv) = g(\mathbf{B}, \mathbf{E}, \rho_m, T, \alpha)$$

Operational context: Determines localized cell probability based on surrounding fluid matrix
Primary simulation reference: SIM-PLASMA-02

EQ-017 — Effective Rotation Modulus Adjustment

$$\Omega_{eff} = \Omega_0 \cdot (1 + \beta T + \gamma \alpha)$$

Operational context: Alters angular momentum transport inside extreme conducting mediums
Primary simulation reference: SIM-PLASMA-02

EQ-018 — Mean Coalition Stopping Power Matrix

$$\left\langle \frac{dE}{dx} \right\rangle_{coalition}$$

Operational context: Tracks spatial energy dissipation of accelerating charges in field medium
Primary simulation reference: SIM-PLASMA-02

EQ-019 — Lepton-Hadron Stopping Power Differential

$$\Delta_{\mu,e,p} = \left\langle \frac{dE}{dx} \right\rangle_{\mu} - \left\langle \frac{dE}{dx} \right\rangle_{e,p}$$

Operational context: Measures energy extraction divergence between muons and light leptons
Primary simulation reference: SIM-PLASMA-02

EQ-020 — Multi-Phase Self-Interaction Effective Potential

$$V(\phi) = \tfrac{1}{2} m_\phi^2 \phi^2 + \lambda \phi^4 + \eta \phi T$$

Operational context: Models stable vacuum expectations when coupled to local Transmissivity
Primary simulation reference: SIM-PLASMA-02

EQ-021 — Scalar Overlay Field Effective Density

$$\rho_{overlay} = \frac{1}{2}\dot{\phi}^2 + V(\phi)$$

Operational context: Represents the scalar stress-tensor contribution to local medium metrics
Primary simulation reference: SIM-PLASMA-02

EQ-022 — Unified Coalition Energy Relation

$$E = \frac{m}{\mu \cdot \varepsilon} \cdot \Xi(T, \alpha, \mathbf{B}, \mathbf{E})$$

Operational context: Modifies effective energy based on local spatial permeability and permittivity
Primary simulation reference: SIM-CORE

EQ-023 — Classical Rest-Mass Energy Bridge

$$E_{legacy} = mc^2$$

Operational context: The standard limit where local dielectric multipliers go to unity
Primary simulation reference: SIM-CORE

EQ-024 — Structural Phase Evolution Vector

$$\frac{d\theta}{dt} = \Upsilon(T, \alpha, \rho_E, \rho_m)$$

Operational context: Tracks state adjustments across multi-phase boundaries
Primary simulation reference: SIM-CORE

EQ-025 — Discrete State Space Constraints

$$\theta \in \{\theta_{init}, \theta_{growth}, \theta_{sat}, \theta_{breach}, \theta_{rec}\}$$

Operational context: Defines permissible phase identities for containment tracking
Primary simulation reference: SIM-CORE

EQ-026 — Standard Gravitational Redshift Identity

$$\frac{\Delta f}{f} = \frac{\Delta \Phi}{c^2}$$

Operational context: Classical control mapping fractional frequency loss over a potential change
Primary simulation reference: SIM-REDSHIFT

EQ-027 — Native Non-Ballistic Photon Momentum Frequency Shift

$$\Delta f = \frac{\Delta p_\gamma}{h}$$

Operational context: Direct relationship linking observed frequency drops to photon momentum depletion
Primary simulation reference: SIM-REDSHIFT

EQ-028 — Integrated Line-Element Gravitational Resistance Vector

$$\Delta p_\gamma = \int g(r)\, dr$$

Operational context: Sums continuous spatial path resistance acting against outward light travel
Primary simulation reference: SIM-REDSHIFT

EQ-029 — Deflection Angular Vector Mapping Integral

$$\Delta\theta = \int \frac{g_\perp}{p_\gamma}\, ds$$

Operational context: Calculates localized cross-field trajectory bending of non-ballistic paths
Primary simulation reference: SIM-REDSHIFT

EQ-030 — Classical Vacuum Asymptotic Light Bending

$$\Delta\theta = \frac{4GM}{c^2 b}$$

Operational context: Standard General Relativity control metric using impact parameter b
Primary simulation reference: SIM-REDSHIFT

EQ-031 — Composite Observed Cosmological Redshift Formula

$$z_{obs} = z_{Doppler} + z_{grav}$$

Operational context: Sums kinematic velocity stretching and native potential path resistance
Primary simulation reference: SIM-REDSHIFT

EQ-032 — Pure Momentum Depletion Fractional Parameter

$$z_{grav} = \frac{\Delta p_\gamma}{p_\gamma}$$

Operational context: Expresses potential-induced redshift directly via loss of photon inertia
Primary simulation reference: SIM-REDSHIFT

EQ-033 — Definitive Non-Singular Collapse Halt Radius

$$r_h = r_s \cdot (1 + \Delta_{EOS})$$

Operational context: Establishes structural arrest boundary scaling past classical Schwarzschild radius
Primary simulation reference: SIM-COLL-03

EQ-034 — Internal Collapse Zone Energy Gradient Flattens

$$\frac{dE}{dr} \to 0$$

Operational context: Vanishing internal work gradient marking non-singular density arrest
Primary simulation reference: SIM-COLL-03

EQ-035 — Local Magnetic Zeeman Spin State Splitting Energy

$$\Delta E = \mu_B \cdot B \cdot m_s$$

Operational context: Calculates shift in microstate orientation potential under high magnetic field
Primary simulation reference: SIM-SPIN-01

EQ-036 — Boltzmann State Occupancy Ratio Profile

$$\frac{P_{up}}{P_{down}} = \exp\left(-\frac{\Delta E}{kT}\right)$$

Operational context: Determines thermal balance of quantum spin alignments in compressed fluid
Primary simulation reference: SIM-SPIN-01

EQ-037 — Absolute Simulation Numerical Precision Metric Floor

$$h_{min} = 10^{-21}$$

Operational context: Prevents floating point roundoff errors during spatial cell collapse
Primary simulation reference: SIM-COLL-03

EQ-038 — Integrated Line-of-Sight Spectral Column Density

$$S(f) = \int \rho(f, r)\, dr$$

Operational context: Calculates net absorption depth for rays escaping internal dense core
Primary simulation reference: SIM-COLL-03

EQ-039 — Standard External Gravitational Time Dilation Curve

$$dt_{obs} = dt_{ref} \cdot \sqrt{1 - \frac{2GM}{rc^2}}$$

Operational context: Classical control mapping coordinate clock slow-down outside mass core
Primary simulation reference: SIM-COLL-03

EQ-040 — Internal Safe Coordinate Time Alignment Vector

$$\frac{dt_{obs}}{dt_{ref}} \approx 1$$

Operational context: Enforces regular physical time movement inside non-singular arrest zone
Primary simulation reference: SIM-COLL-03

EQ-041 — Standard Clock Shift Metric Projection

$$f_{obs} = f_{emit} \cdot \left(1 + \frac{\Delta\Phi}{c^2}\right)$$

Operational context: Relates emitted frequencies to observed frames across potential steps
Primary simulation reference: SIM-COLL-03

EQ-042 — Classical Newtonian Kinematic Linear Velocity

$$v = v_0 + a \cdot t$$

Operational context: Standard flat-space linear velocity under constant acceleration
Primary simulation reference: SIM-CORE

EQ-043 — Newtonian Time Duration Interval Solver

$$t = \frac{v - v_0}{a}$$

Operational context: Extracts time duration required to reach targeted velocity thresholds
Primary simulation reference: SIM-CORE

EQ-044 — Temporal Unit Conversion Vector Scalar

$$textdays = \frac{t}{86400}$$

Operational context: Normalizes metric simulation seconds into real calendar days
Primary simulation reference: SIM-CORE

EQ-045 — Relativistic Flat Velocity Coordinate Asymptote

$$v(t) = \frac{a \cdot t}{\sqrt{1 + \left(\frac{a \cdot t}{c}\right)^2}}$$

Operational context: Shows linear coordinate time approach to speed of light c
Primary simulation reference: SIM-CORE

EQ-046 — Proper-Time Relativistic Acceleration Vector Form

$$v(\tau) = c \cdot \tanh\left(\frac{a \cdot \tau}{c}\right)$$

Operational context: Tracks experienced system timeline acceleration mapping to hyperbolic velocity
Primary simulation reference: SIM-CORE

EQ-047 — Standard Mass-Energy Equivalence Baseline

$$E = m \cdot c^2$$

Operational context: Classical rest energy baseline for standard reference fields
Primary simulation reference: SIM-CORE

EQ-048 — Dynamic Mass-Equivalent Gravitational Energy Conversion

$$m_{equiv} = \frac{E}{c^2}$$

Operational context: Translates pure electromagnetic or scalar field energy into effective mass
Primary simulation reference: SIM-CORE

EQ-049 — Planck-Einstein Photon Energy Quantum Identity

$$E = h \cdot \nu$$

Operational context: Establishes linear relationship between frequency and quantum energy packets
Primary simulation reference: SIM-CORE

EQ-050 — Standard Linear Photon Momentum Relation

$$p = \frac{E}{c}$$

Operational context: Direct link relating electromagnetic wave packet energy to pure inertia
Primary simulation reference: SIM-CORE

EQ-051 — Effective Photon Gravitational Mass Identity

$$m_{equiv} = \frac{h \cdot \nu}{c^2}$$

Operational context: Calculates instant gravitational mass equivalence of localized light rays
Primary simulation reference: SIM-CORE

EQ-052 — Simplified Non-Permeable Base Medium Energy Scalar

$$E = \frac{m}{\mu \cdot \varepsilon}$$

Operational context: Alters effective baseline inertia based on spatial index conditions
Primary simulation reference: SIM-FLUID-01

EQ-053 — Classical Linear Momentum Conservation Identity

$$p = m \cdot v$$

Operational context: Tracks mass translation vectors across fluid boundary cells
Primary simulation reference: SIM-FLUID-01

EQ-054 — Macroscopic Angular Momentum Tracking Array

$$L = I \cdot \omega$$

Operational context: Measures rotational stability of expanding or collapsing accretion disks
Primary simulation reference: SIM-FLUID-01

EQ-055 — Geometric Moment of Inertia Structural Definition

$$I = k \cdot M \cdot R^2$$

Operational context: Applies structural coefficients k to model mass distribution geometries
Primary simulation reference: SIM-FLUID-01

EQ-056 — Classical Hydrostatic Spatial Equilibrium Gradient

$$\frac{dP}{dr} = -\rho \cdot g$$

Operational context: Balances internal isotropic fluid pressure against downward gravity forces
Primary simulation reference: SIM-FLUID-01

EQ-057 — Macroscopic Bulk Mean Mass Density

$$\rho = \frac{M}{V}$$

Operational context: Simple ratio tracking structural volume filling thresholds
Primary simulation reference: SIM-FLUID-01

EQ-058 — Isotropic Mechanical Pressure Force Projection

$$P = \frac{F}{A}$$

Operational context: Measures perpendicular mechanical force distributed per area unit
Primary simulation reference: SIM-FLUID-01

EQ-059 — Parabolic Stellar Density Radial Profile Matrix

$$\rho(r) = \rho_0 \cdot \left[1 - \left(\frac{r}{R}\right)^2\right]$$

Operational context: Standard non-uniform interior density tracking function for mass cores
Primary simulation reference: SIM-FLUID-01

EQ-060 — Enclosed Radial Mass Volume Profiler Integral

$$M(r) = \int 4\pi r^2 \rho(r)\, dr$$

Operational context: Integrates nested shells to find absolute structural mass up to radius r
Primary simulation reference: SIM-FLUID-01

EQ-061 — Localized Newton Gravitational Field Acceleration

$$g(r) = \frac{G \cdot M(r)}{r^2}$$

Operational context: Determines inward gravitational force pulling on interior fluid packets
Primary simulation reference: SIM-FLUID-01

EQ-062 — Boundary Surface Gravity Acceleration Endpoint

$$g_{surface} = \frac{G \cdot M}{R^2}$$

Operational context: Calculates gravitational grip at the absolute edge of active core mass
Primary simulation reference: SIM-FLUID-01

EQ-063 — Uniform Spherical Gravitational Binding Potential Energy

$$U = -\frac{3GM^2}{5R}$$

Operational context: Classical benchmark tracking total work required to dismantle core structure
Primary simulation reference: SIM-FLUID-01

EQ-064 — Total Standard Mechanical Conserved Energy Scalar

$$E_{tot} = K + U$$

Operational context: Monitors sum of macroscopic kinetic translation and potential pools
Primary simulation reference: SIM-FLUID-01

EQ-065 — Fluid Mass Continuity Matrix

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

Operational context: Enforces absolute mass conservation across localized dynamic fluid updates
Primary simulation reference: SIM-FLUID-01

EQ-066 — Navier-Stokes Uncharged Pure Fluid Momentum Axis

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right) = -\nabla P$$

Operational context: Tracks mechanical pressure gradient accelerations without electromagnetic forces
Primary simulation reference: SIM-FLUID-01

EQ-067 — Ideal Gas Thermodynamic Equation of State

$$P = \rho \cdot R_{specific} \cdot T_k$$

Operational context: Couples bulk thermal kinetic temperature to mechanical gas pressures
Primary simulation reference: SIM-FLUID-01

EQ-068 — Macroscopic Rotational Kinetic Energy Scalar

$$E_{rot} = \frac{1}{2} I \cdot \omega^2$$

Operational context: Measures kinetic track energy stored within spinning stellar geometries
Primary simulation reference: SIM-FLUID-01

EQ-069 — Linear Equatorial Local Tangential Velocity

$$v_{rot} = \omega \cdot r$$

Operational context: Tracks rotational speed of boundary cells at radius offsets
Primary simulation reference: SIM-FLUID-01

EQ-070 — Classical Magnetic Dipole Radial Falloff Vector

$$B(r) = B_0 \cdot \left(\frac{R}{r}\right)^3$$

Operational context: Models decay of global stellar fields moving away from active core poles
Primary simulation reference: SIM-MAG-02

EQ-071 — Localized Magnetic Field Energy Storage Density

$$u_B = \frac{B^2}{2\mu_0}$$

Operational context: Tracks spatial energy density bound within magnetic flux structures
Primary simulation reference: SIM-MAG-02

EQ-072 — Isotropic Magnetic Field Outward Counterforce Pressure

$$P_B = \frac{B^2}{2\mu_0}$$

Operational context: Electromagnetic pressure vector acting to arrest uncharged collapse trajectories
Primary simulation reference: SIM-MAG-02

EQ-073 — Microscopic Charged Particle Lorentz Force Equation

$$F_Lorentz = q \cdot (\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Operational context: Defines path acceleration vectors for individual free plasma charges
Primary simulation reference: SIM-MAG-02

EQ-074 — Net Magnetic Flux Surface Integrator Matrix

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

Operational context: Measures total magnetic field lines piercing a given reference boundary
Primary simulation reference: SIM-MAG-02

EQ-075 — Core Native Gradient Reversal Threshold Boundary

$$\lim_{r \to r_h^+} \frac{dE}{dr} = 0$$

Operational context: Arrests internal collapse work perfectly as boundary moves to Halt Radius
Primary simulation reference: SIM-COLL-03

EQ-076 — Dynamic Plasma Escape Frequency Cutoff

$$\omega_\gamma > \omega_p \cdot \Theta(T, \mathbf{B}, \rho_m)$$

Operational context: Determines threshold where photon signals slice through shielding media
Primary simulation reference: SIM-PLASMA-01

EQ-077 — Effective Modified Plasma Transmissivity Diffusion Matrix

$$D_{eff} = D_0 \cdot \Psi(T, \alpha, \rho_m, \mathbf{B})$$

Operational context: Controls charge transport behavior across intense local phase shifts
Primary simulation reference: SIM-PLASMA-01

EQ-078 — Modified Unified Spatial Fick Concentration Flux Vector

$$\mathbf{J} = -D_{eff} \nabla n$$

Operational context: Governs fluid element diffusion vectors inside heavy compression zones
Primary simulation reference: SIM-PLASMA-01

EQ-079 — The Definitive Trinca Maximum Mass Density Ceiling

$$\rho_{max} = textConstant$$

Operational context: Absolute cosmological maximum density allowed by universal phase architecture
Primary simulation reference: SIM-COLL-03

EQ-080 — Unified Metric Spacetime Reality Function

$$\mathcal{R}(x) = f(textenergy, textcurvature, textmedium)$$

Operational context: Universal master tracking trait monitoring absolute metric status parameters
Primary simulation reference: SIM-CORE

EQ-081 — Lepton Rest-Energy Quantum Frequency Identity

$$E_0 = m_e c^2 = h f_c$$

Operational context: Bridges rest-mass inertia directly to its intrinsic quantum clock speed
Primary simulation reference: SIM-SPIN-01

EQ-082 — Electron Intrinsic Compton Clock Frequency Baseline

$$f_c = \frac{m_e c^2}{h}$$

Operational context: Establishes fundamental quantum oscillation frequency (~1.236e20 Hz)
Primary simulation reference: SIM-SPIN-01

EQ-083 — Macroscopic Maxwell Gauss Electrostatic Core

$$\nabla \cdot \mathbf{E} = \frac{\rho_e}{\varepsilon_0}$$

Operational context: Maps total electric displacement field divergence directly to free charges
Primary simulation reference: SIM-MAG-02

EQ-084 — Maxwell Gauss Magnetic Non-Monopole Constraint

$$\nabla \cdot \mathbf{B} = 0$$

Operational context: Enforces continuous divergence-free closed tracking loops for all magnetic lines
Primary simulation reference: SIM-MAG-02

EQ-085 — Maxwell Faraday Time-Varying Field Induction Matrix

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Operational context: Calculates electrical electromotive curls induced by moving magnetic boundaries
Primary simulation reference: SIM-PLASMA-01

EQ-086 — Maxwell Ampere Extended Circuital Current Dynamic Loop

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

Operational context: Relates magnetic curls to real currents and displacement field changes
Primary simulation reference: SIM-MAG-02

EQ-087 — Coupled MHD Navier-Stokes Lorentz Fluid Momentum Equation

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right) = -\nabla P + \mathbf{J} \times \mathbf{B} + \rho \mathbf{g}$$

Operational context: Integrates mechanical pressure, gravitational grip, and macroscopic magnetic braking
Primary simulation reference: SIM-FLUID-01

EQ-088 — Ideal Fluid MHD Magnetic Flux-Freezing Induction Identity

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B})$$

Operational context: Locks magnetic lines securely into the translation path of perfectly conducting plasma
Primary simulation reference: SIM-PLASMA-02

EQ-089 — Continuous Dynamic MHD Divergence-Free Constraint

$$\nabla \cdot \mathbf{B}_{fluid} = 0$$

Operational context: Maintains clean geometric fields across grid mesh translations to prevent simulation errors
Primary simulation reference: SIM-FLUID-01

EQ-090 — Total Volume Integrated Magnetohydrodynamic Magnetic Reservoir Pool

$$E_B = \int \frac{B^2}{2\mu_0}\, dV$$

Operational context: Sums absolute interior magnetic field energy supporting collapsing mass scales
Primary simulation reference: SIM-COLL-03

EQ-091 — Triple Spatial Volume Mass Integration Equation

$$\text{Mass}_{enclosed} = \iiint_V \rho(r, \theta, \phi)\, dV$$

Operational context: Sums dynamic density variations across spherical shells to extract total true mass
Primary simulation reference: SIM-FLUID-01

EQ-092 — Triple Volume Electrostatic Charge Allocation Integrator

$$\Phi_{total} = \iiint_V \rho_e\, dV$$

Operational context: Calculates net interior charge bound within three-dimensional cell matrices
Primary simulation reference: SIM-MAG-02

EQ-093A — Boundary Limit Horizon Convergence Law

$$\lim_{r \to r_h^+} \mathcal{R}(x) = \mathcal{R}_{critical}$$

Operational context: Determines metric parameter values at the exact spatial boundary interface of the Halt Radius.
Primary simulation reference: SIM-COLL-03

EQ-093B — Inflation Replacement Continuous Field Decay Rule

$$v_{collapse}(r) = c \cdot \left(1 - \frac{r_h}{r}\right)^{\gamma_{medium}}$$

Operational context: Models velocity dampening as a function of medium density profiles to prevent infinite singularities.
Primary simulation reference: SIM-MEDIA-01

EQ-094 — Standard Time-Dependent Schrodinger Wave Equation Control Group

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r},t)\right]\Psi(\mathbf{r},t)$$

Operational context: Quantum baseline control tracking wave function evolution under external potentials
Primary simulation reference: SIM-SPIN-02

EQ-095 — Lenz's Law Electromagnetic Braking Conservation Induction Rule

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Operational context: Calculates backward EMF counter-torques acting to stabilize plasma accretion paths
Primary simulation reference: SIM-PLASMA-01

EQ-096 — The Tri-Factor Frequency Dilation Law

$$\Delta f = -\left(\frac{v^2}{2c^2}\right) f_0 - \left(\frac{\Delta\Phi}{c^2}\right) f_0 - \kappa_a a$$

Operational context: Proves clock shifts are a tripartite function of kinematic velocity, localized spatial potential, and proper acceleration vector interactions.
Primary simulation reference: SIM-SPIN-02

EQ-097 — Vector Signal Geometry Phase Arrival Matrix

$$f_{\text{obs}}(\mathbf{r},\theta) = f_{\text{emit}}\left(1 - \frac{\mathbf{v}\cdot\hat{\mathbf{r}}}{c} - \frac{a\cdot\mathbf{r}}{c^2}\cos\theta\right)$$

Operational context: Models directional and distance-dependent wavefront arrival rates on a coordinate detector grid.
Primary simulation reference: SIM-SPIN-02

EQ-098 — Medium-Law Refractive Gradient Lensing Equation

$$\alpha = \int_{-\infty}^{\infty} \frac{1}{n(r)} \left(\nabla n(r) \cdot \hat{\mathbf{n}}\right) ds$$

Operational context: Replaces geometric spacetime bending by modeling light deflection as pure optical refraction through a variable vacuum matrix.
Primary simulation reference: SIM-MEDIA-01

EQ-099 — Frequency-Radius Relation Exponential Decay

$$\nu(r) = \nu_0 \cdot \exp\left(-\int_{r_0}^{r} \mu_{\text{vac}}(s)\rho_{\text{med}}(s)\, ds\right)$$

Operational context: Tracks cumulative transmission path phase changes and frequency modifications over varying medium density profiles.
Primary simulation reference: SIM-MEDIA-01

EQ-100 — Constitutive Vacuum Wave-Vector Formulation

$$\mathbf{k}_{\text{local}} = \omega\sqrt{\varepsilon_{\text{local}}\mu_{\text{local}}}\hat{\mathbf{k}}$$

Operational context: Locks local electromagnetic propagation velocity and direction explicitly to the spatial density profile of the local medium substrate.
Primary simulation reference: SIM-MEDIA-01

EQ-101 — Linear Superposition Vector Acceleration Matrix Slice 1

$$\mathbf{a}_{101} = -\sum_{j \neq i} \nabla\Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-102 — Linear Superposition Vector Acceleration Matrix Slice 2

$$\mathbf{a}_{102} = -\sum_{j \neq i} \nabla\Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-103 — Linear Superposition Vector Acceleration Matrix Slice 3

$$\mathbf{a}_{103} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-104 — Linear Superposition Vector Acceleration Matrix Slice 4

$$\mathbf{a}_{104} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-105 — Linear Superposition Vector Acceleration Matrix Slice 5

$$\mathbf{a}_{105} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-106 — Linear Superposition Vector Acceleration Matrix Slice 6

$$\mathbf{a}_{106} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-107 — Linear Superposition Vector Acceleration Matrix Slice 7

$$\mathbf{a}_{107} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-108 — Linear Superposition Vector Acceleration Matrix Slice 8

$$\mathbf{a}_{108} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-109 — Linear Superposition Vector Acceleration Matrix Slice 9

$$\mathbf{a}_{109} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-110 — Linear Superposition Vector Acceleration Matrix Slice 10

$$\mathbf{a}_{110} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-111 — Linear Superposition Vector Acceleration Matrix Slice 11

$$\mathbf{a}_{111} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-112 — Linear Superposition Vector Acceleration Matrix Slice 12

$$\mathbf{a}_{112} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-113 — Linear Superposition Vector Acceleration Matrix Slice 13

$$\mathbf{a}_{113} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-114 — Linear Superposition Vector Acceleration Matrix Slice 14

$$\mathbf{a}_{114} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-115 — Linear Superposition Vector Acceleration Matrix Slice 15

$$\mathbf{a}_{115} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-116 — Linear Superposition Vector Acceleration Matrix Slice 16

$$\mathbf{a}_{116} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-117 — Linear Superposition Vector Acceleration Matrix Slice 17

$$\mathbf{a}_{117} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-118 — Linear Superposition Vector Acceleration Matrix Slice 18

$$\mathbf{a}_{118} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-119 — Linear Superposition Vector Acceleration Matrix Slice 19

$$\mathbf{a}_{119} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

EQ-120 — Linear Superposition Vector Acceleration Matrix Slice 20

$$\mathbf{a}_{120} = -\sum_{j \neq i} \nabla \Phi_j(\mathbf{r}_i - \mathbf{r}_j)$$

Operational context: Computes high-fidelity multi-body collision-free orbital drift trajectories via direct spatial field summation loops.
Primary simulation reference: SIM-COLL-03

Rendering notes

The following entry contained an OOXML-style carriage-return artifact that prevented mathematical typesetting. The visual rendering substituted the apparent intended TeX command solely so the equation could be displayed; the original ledger string should remain the audit source.

EQ-086
Original ledger string: `\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partia`
`\mathbf{E}}{\partial t}\_x000D_ight)`
Rendered as: `\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial`
`\mathbf{E}}{\partial t}\right)`